PSP Aurora Postgres Database Migration - Summary

Summary

As part of Oracle Exit initiative, PSP databases needs to migrate to Aurora PostgreSQL by FY24. PSP database includes 2 databases i.e. Primary /Monolith, whose size is ~24 TB and Secondary/IBOB database, whose size is ~5 TB.

Proof Of Concept (POC)

Secondary Database

Tasks	Non-prod Completion (%) - (SYS Env)	Prod Completion (%)	Notes
Database Infra Setup	70%	70%	Monitoring work is in progress
Data Migration	COMPLETED	COMPLETED	Replication work is in progress
Data Validation	IN PROGRESS	IN PROGRESS	
Application Readiness	YET TO START	YET TO START	

Primary Database

Tasks	Non-prod Completion (%)	Prod Completion (%)	Notes
Database Infra Setup	50%	50%	Need to work on monitoring
Data Migration	10%	10%	Replication work is in progress
Data Validation	IN PROGRESS	IN PROGRESS	
Application Readiness	YET TO START	YET TO START	

Details

Secondary

Data migration

Tasks	Non-Prod	Prod

Setup Aurora PostgreSQL database. Started using IPS as well for infra build.	COMPLETE	
Convert schema for Aurora PostgreSQL using SCT (AWS Schema Conversion Tool) for migration POC	COMPLETE	
Configure migration tasks to migrate data from Oracle to Aurora PostgreSQL.	COMPLETE	
Data migration (Full data load + Apply cached changes, CDC)	COMPLETE	
Completed data migration POC for secondary database.	COMPLETE	

Data validation

Tasks	Non-Prod	Prod
There were some findings and challenges during this which is listed below in challenges section		

Application Readiness

Tasks/POC	Non-Prod	Prod
Setup locally Aurora PostgreSQL 12.7 and create a wiki		
Work on schema conversion for Secondary database		
Access patterns for Queries on Partitioned tables		

Assumptions

1. Application would have dual writes enabled for whole migration strategy. Hence data replication from Aurora PostgreSQL to Oracle is not worked upon.

Findings

1. SCT does not convert schema properly. There is further manual work required to get objects in sync with oracle.
    a. FK referring to partitioned tables
    b. Check constraints on partitioned tables
2. DMS for data validation
    - Compare non-partition tables without CLOB columns
    - DMS validation scalability for busy tables.
    - Can't compare time range partition tables (as it does not have PK)
    - CLOB columns appearing as OOS most of time
3. Got separate license for validation tool "DB SOLO". Need to start using this for data validation.

Challenges

1. Data migration for CLOB data
2. Data migration for secondary database at production scale
3. Data validation at production scale
4. Identify partitioning mechanism and apply for PSP w.r.t. Aurora PostgreSQL

Blockers

1. Table partitioning which would impact data integrity as well as performance. More details in PSP Data migration
2. Application Schema generation/conversion has not yet started.

Next Steps

1. Application needs to be ready in early Q2, tip-toe needs to be started in early Q3 followed by touch down towards end of Q3.
2. Ensure assumptions are well understood and backed into application design.
3. Team to analyze partitioning challenge and come up with solution.

Monolith

Data migration

Tasks	Status
Setup Aurora PostgreSQL database. Started using IPS as well for infra build.	COMPLETED
Schema Conversion for Aurora PostgreSQL using SCT (AWS Schema Conversion Tool) for migration POC.	10%
Configure migration tasks to migrate data from Oracle to Aurora PostgreSQL.	YET TO START
Data migration (Full data load + Apply cached changes, CDC)	YET TO START
Optimizing for replication performance	YET TO START

Data validation

Topics	Status
There were some findings and challenges during this which is listed below in challenges section	OPEN

Application Readiness

Tasks/POC	Non-Prod	Prod

Assumption

1. Application would have dual writes enabled for whole migration strategy. Hence data replication from Aurora PostgreSQL to Oracle is not worked upon.

Findings

1. SCT does not convert schema properly. There is further manual work required to get objects in sync with oracle.
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Challenges

1. Data migration for CLOB data
2. Data migration for Primary database at production scale
3. Data validation at production scale
4. Identify partitioning mechanism and apply for PSP w.r.t. Aurora PostgreSQL

Blockers

1. Table partitioning which would impact data integrity as well as performance. More details in PSP Data migration
2. Application Schema generation/conversion has not yet started.

Next Steps

1. Application needs to be ready, prepare for tip-toe needs followed by touch down.
2. Ensure assumptions are well understood and backed into application design.
3. Team to analyze partitioning challenge and come up with solution.

Ask to Leadership

1. Had raised request to get PostgreSQL contractor resource for DBE but we haven't got yet.